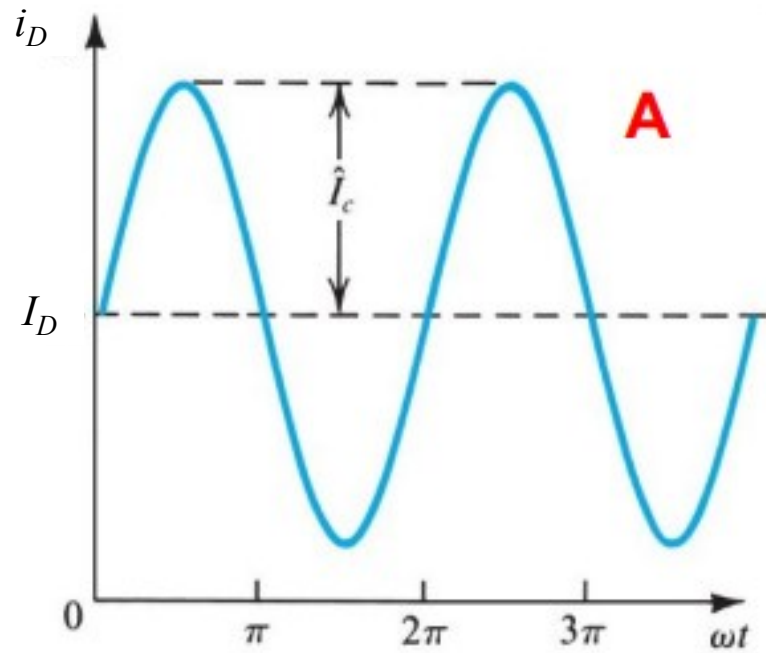
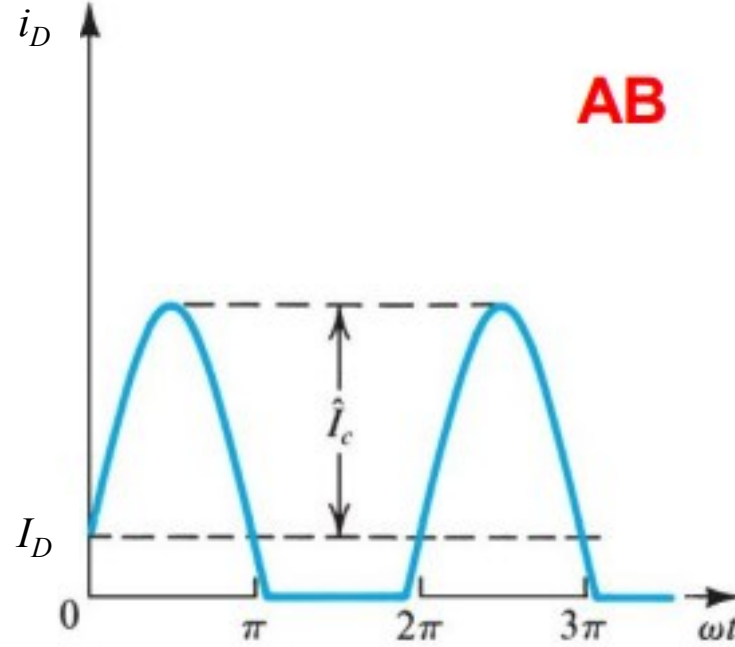


OUTPUT STAGES

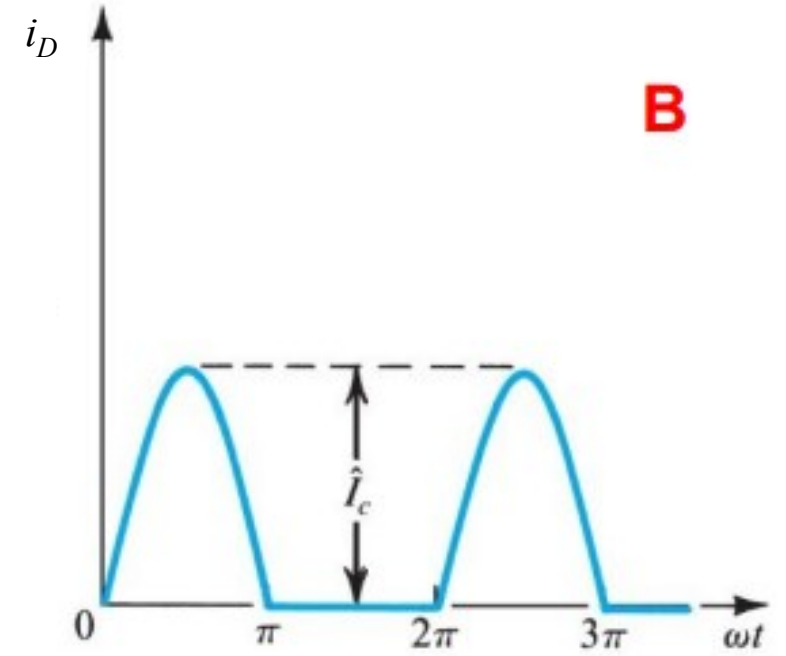
OUTPUT STAGE TYPES



Class A

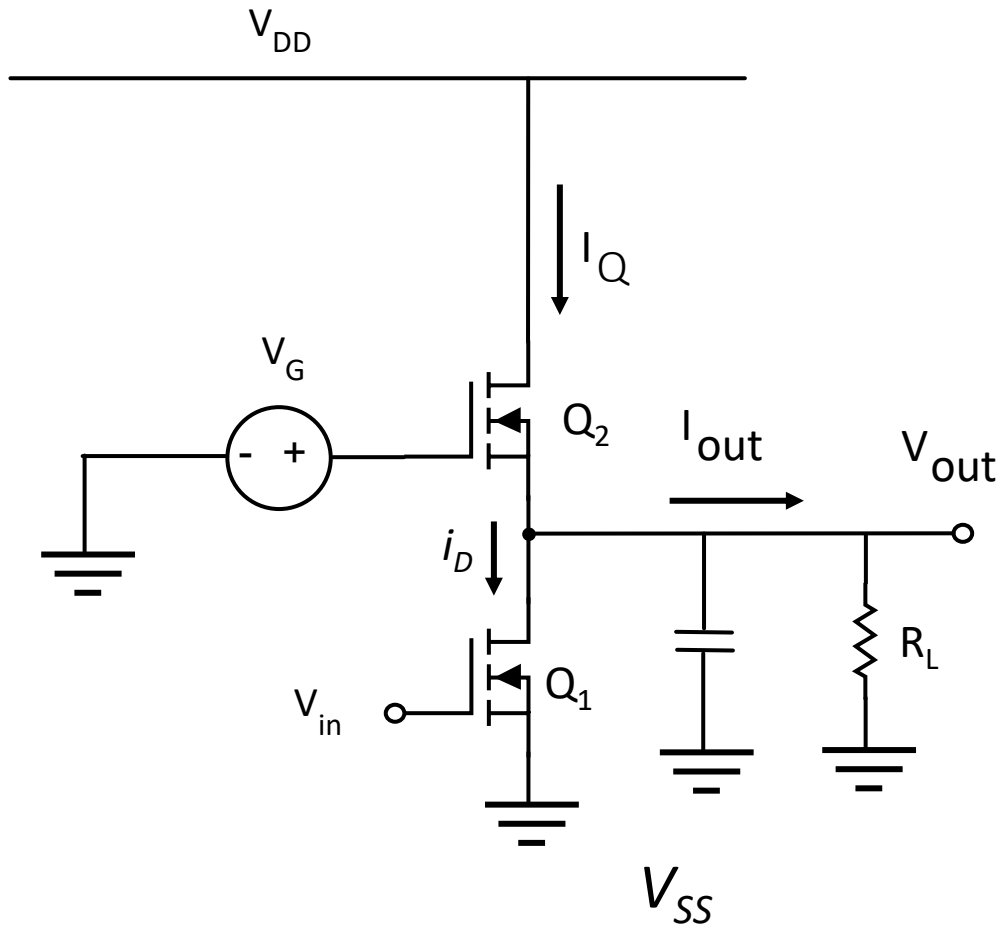


Class AB



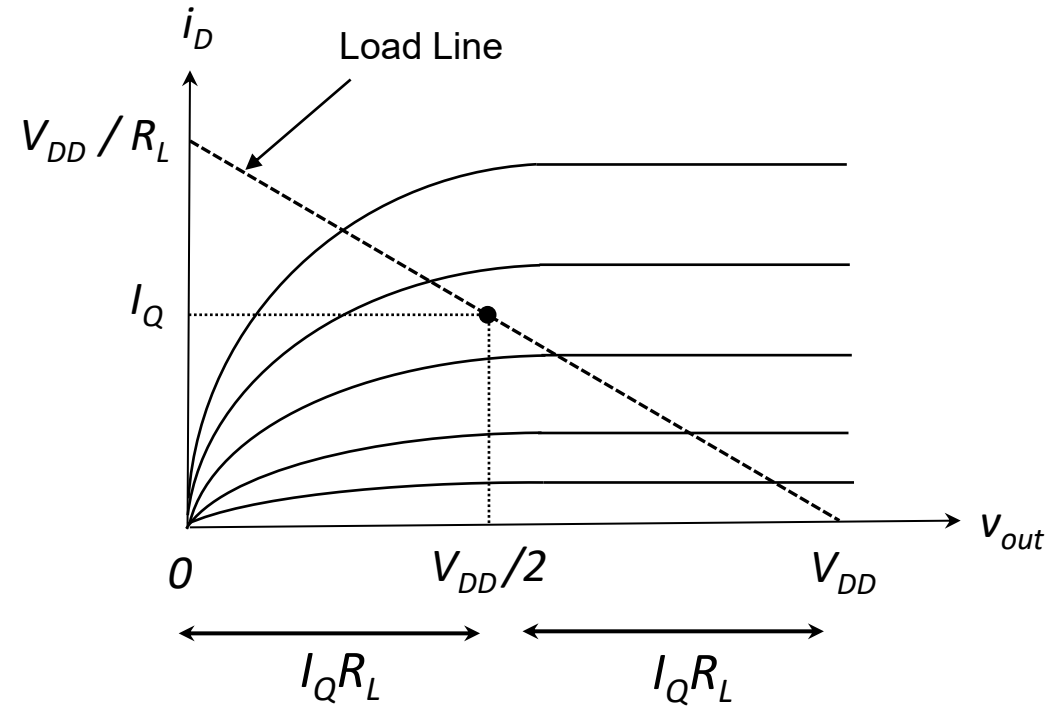
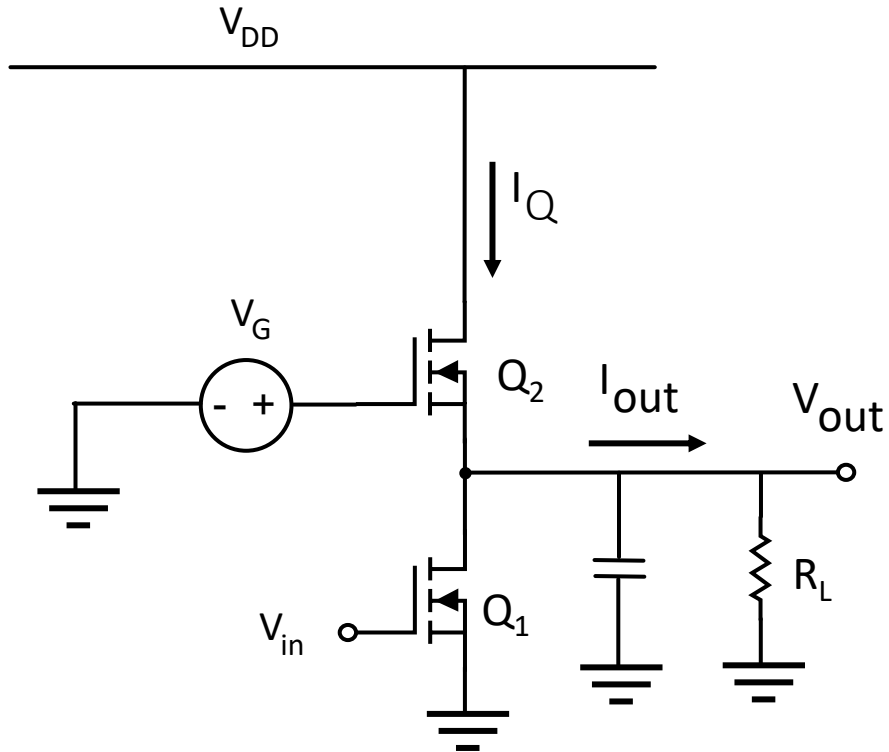
Class B

CLASS A COMMON SOURCE OUTPUT STAGE



- Poor Efficiency
- Class A output stage transistor conduct current over an entire input cycle
- Linear

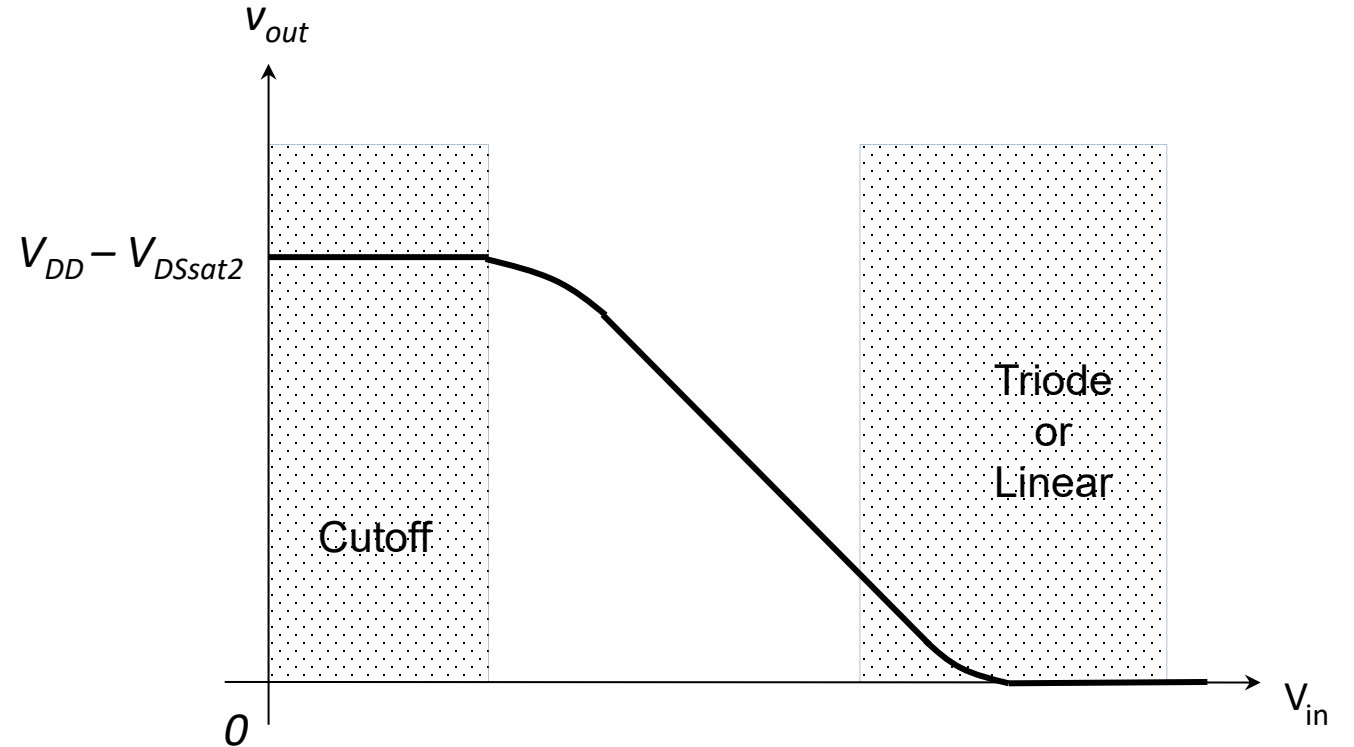
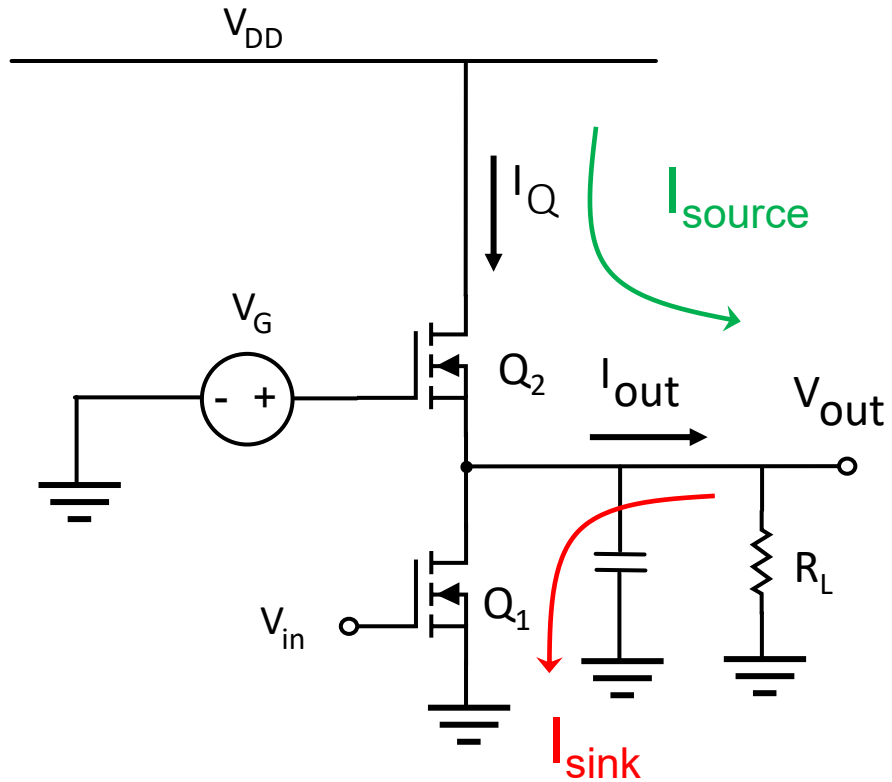
CLASS A COMMON SOURCE OUTPUT STAGE



$$Efficiency = \frac{P_{RL}}{P_{Supply}} = \frac{\frac{v_{out(peak)}^2}{2R_L}}{(V_{DD})I_Q} = \frac{\frac{v_{out(peak)}^2}{2R_L}}{(V_{DD})\frac{(V_{DD})}{2R_L}} = \left(\frac{v_{out(peak)}}{V_{DD}} \right)^2 \quad v_{rms} = v_{out(peak)}/\sqrt{2}$$

Maximum efficiency occurs when $v_{OUT(peak)} = V_{DD} / 2$ which gives 25%.

CLASS A COMMON SOURCE OUTPUT STAGE



$$I_{sink} = \frac{\mu_n C_{ox} W_1}{2L_1} (V_{in} - V_{th1})^2$$

$$I_{source} = \frac{\mu_n C_{ox} W_2}{2L_2} (V_G - V_{out} - V_{th2})^2$$

$$R_{out} = \frac{1}{g_{ds1} + g_{ds2}}$$

$$I_{out} = I_{source} - I_{sink}$$

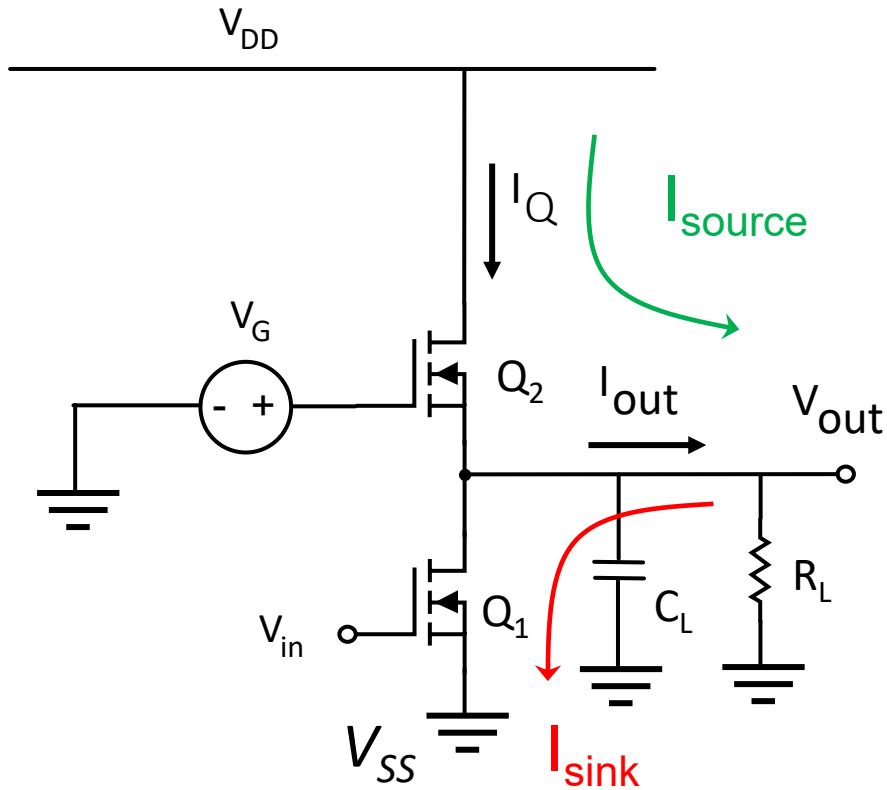
CLASS A COMMON SOURCE OUTPUT STAGE

We want to improve the Slew Rate

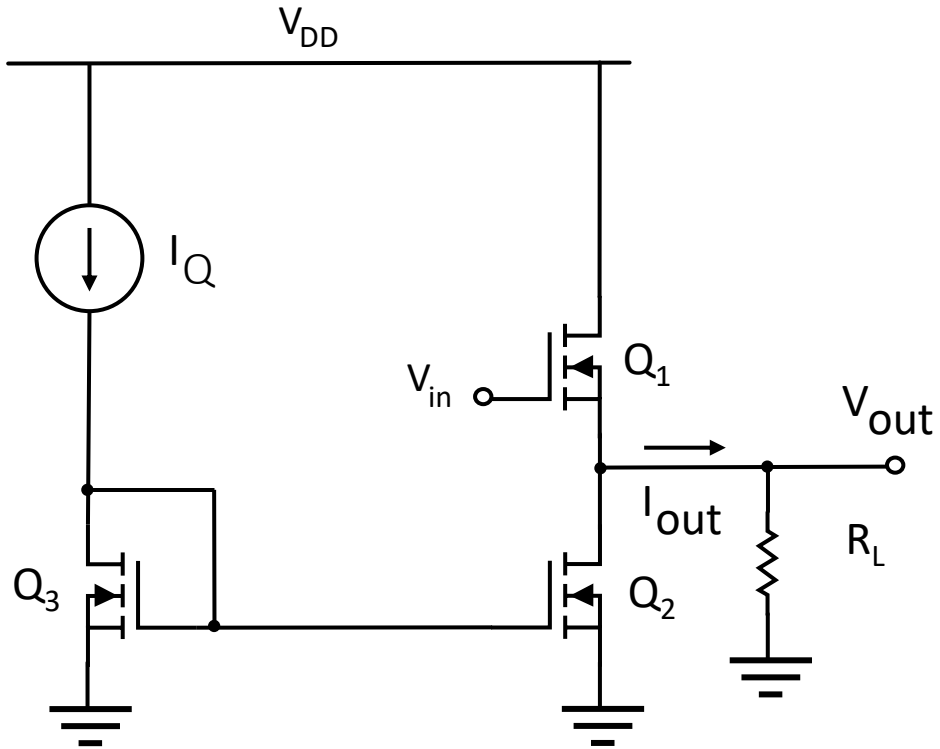
$$|I_{out}| > C_L \cdot \text{Slew Rate}$$

$$|I_{out}| > \frac{V_{out}(\text{peak})}{R_L}$$

The maximum current will be determined by both the current required to provide the necessary slew rate (C_L) and the current required to provide a voltage across the load resistor (R_L).



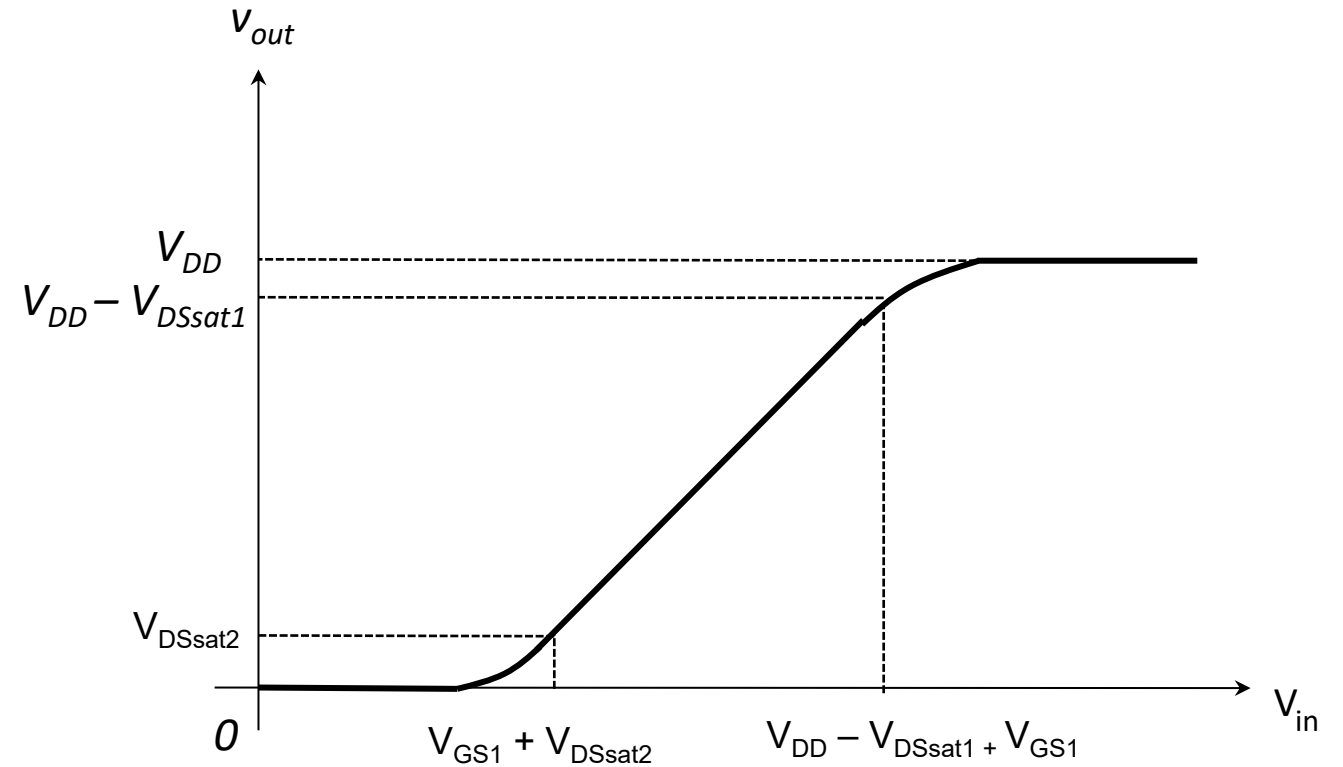
CLASS A SOURCE FOLLOWER OUTPUT STAGE



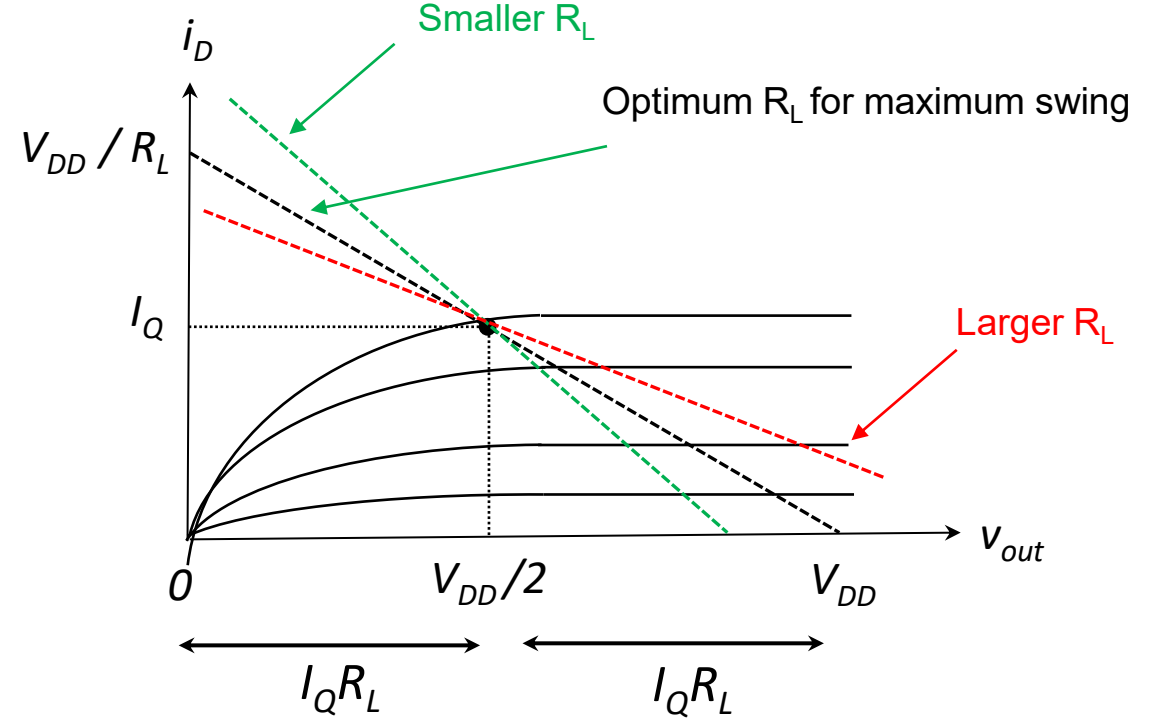
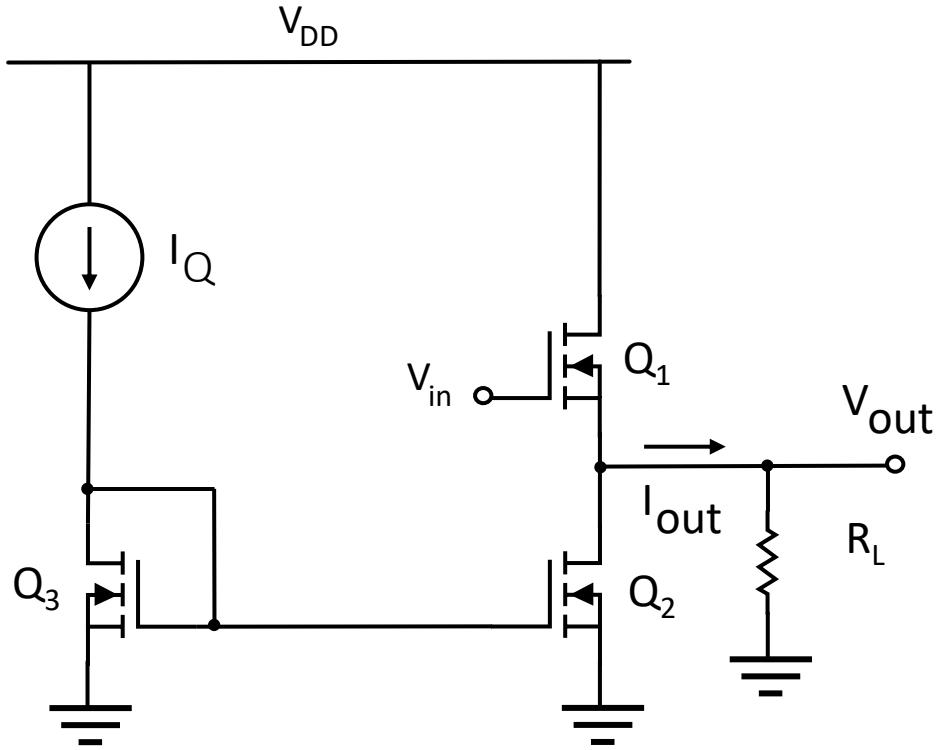
$$V_{DSsat1} = V_{GS1} - V_{th1}$$

$$V_{DSsat2} = V_{GS2} - V_{th2}$$

Voltage Transfer Curve and Maximum Output Voltage Swings



CLASS A SOURCE FOLLOWER OUTPUT STAGE



Maximum efficiency occurs for the optimum value of R_L which gives maximum swing.

$$Efficiency = \frac{P_{RL}}{P_{Supply}} = \frac{\frac{v_{out(peak)}^2}{2R_L}}{(V_{DD})I_Q} = \frac{\frac{v_{out(peak)}^2}{2R_L}}{(V_{DD})\frac{(V_{DD})}{2R_L}} = \left(\frac{v_{out(peak)}}{V_{DD}} \right)^2$$

CLASS AB AND CLASS B OUTPUT STAGE

- Push-Pull Source Follower
- Push-Pull Common-Source Amplifier

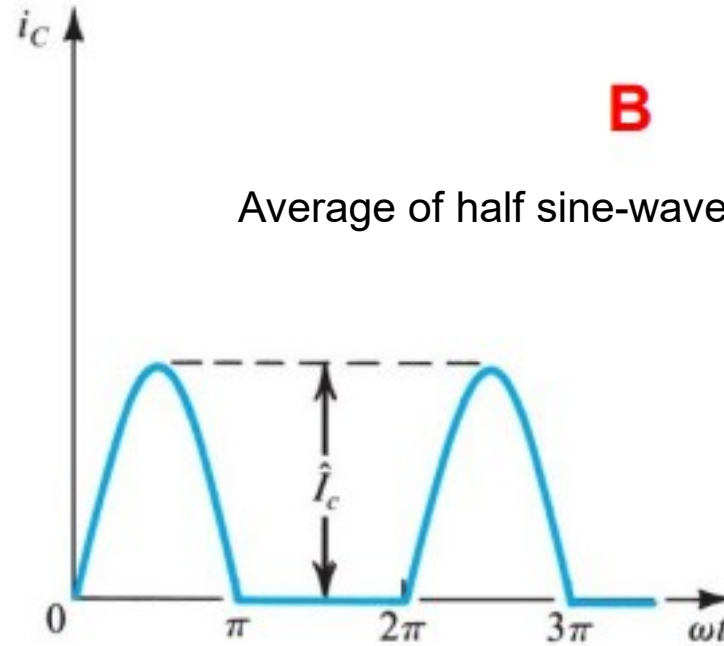
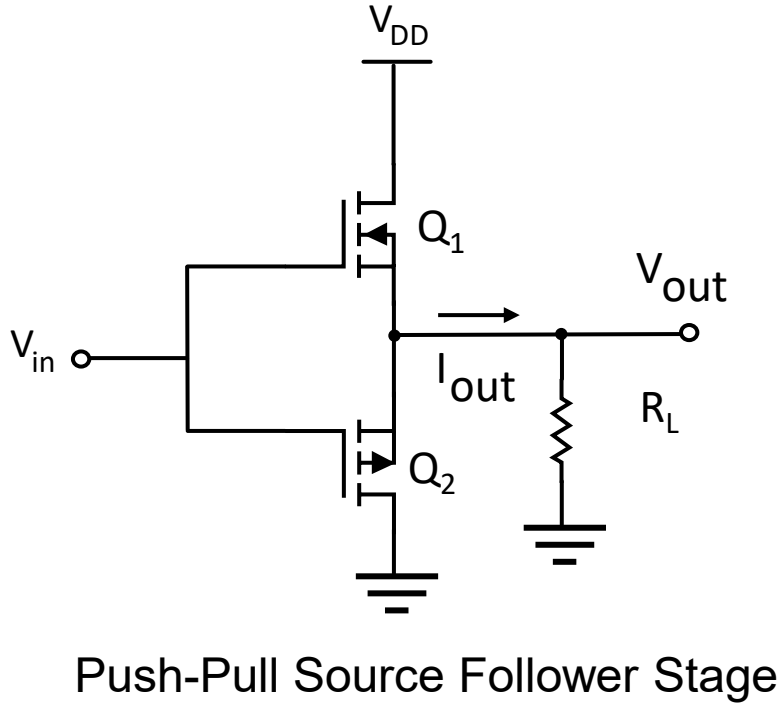
Efficiency depends on how the transistors are biased.

- Class B - one transistor has current flow for only 180° of the sinusoid (half period).
- Class AB - each transistor has current flow for more than 180° of the sinusoid.

Maximum efficiency is between 25% and 78.5%.

CLASS B OUTPUT STAGE

- Class B output stages have only one transistor conducting current for each half cycle



$$i_{out} = \frac{v_{out(peak)}}{R_L} \sin(\omega t)$$

$$I_{DD} = \frac{1}{2\pi} \int_0^\pi I_{peak} \sin\theta d\theta = \frac{2}{2\pi} I_{peak}$$

$$I_{peak} = \frac{v_{out(peak)}}{R_L}$$

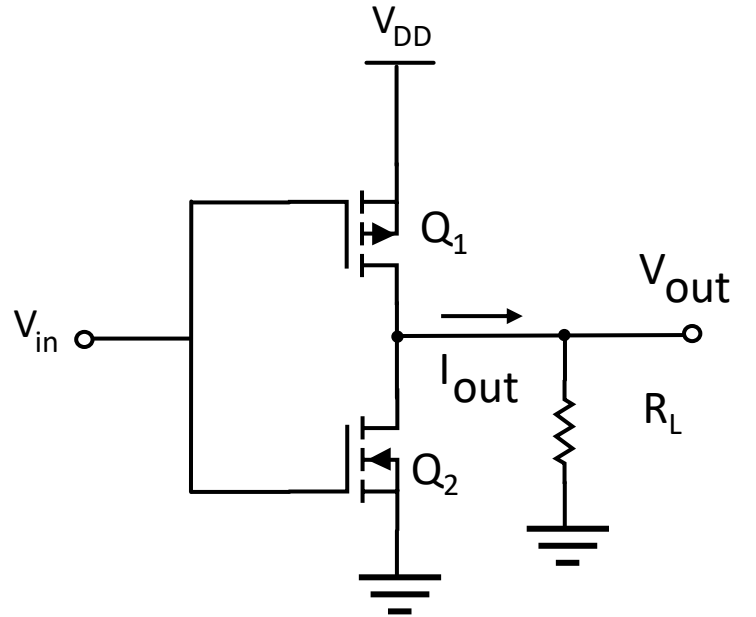
$$I_{DD} = \frac{2v_{out(peak)}}{2\pi R_L}$$

$$Efficiency = \frac{P_{RL}}{P_{VDD}} = \frac{\frac{v_{out(peak)}^2}{2R_L}}{(V_{DD}) \left(\frac{2v_{out(peak)}}{2\pi R_L} \right)} = \frac{\pi v_{out(peak)}}{2 V_{DD}}$$

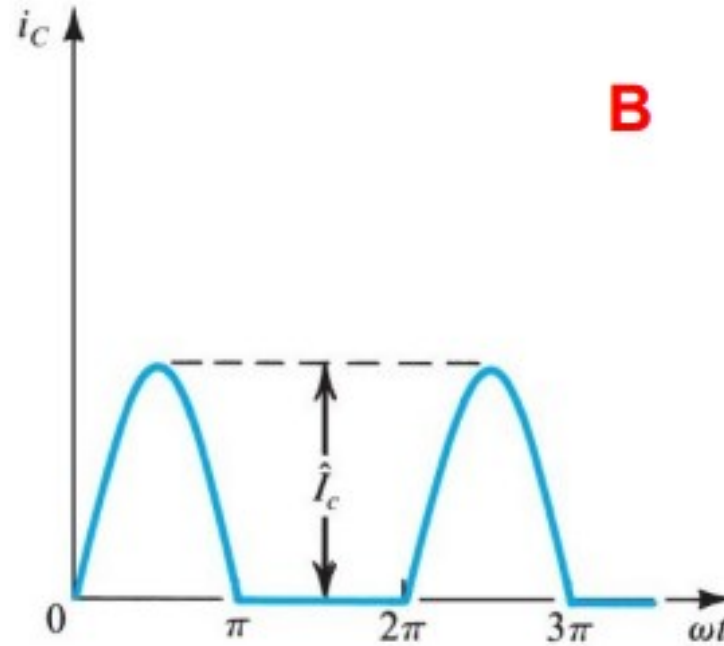
If $V_{out} = V_{DD}/2$, then Efficiency = 78.5%

CLASS B OUTPUT STAGE

- Class B output stages have only one transistor conducting current for each half cycle



Push-Pull Common-Source Stage

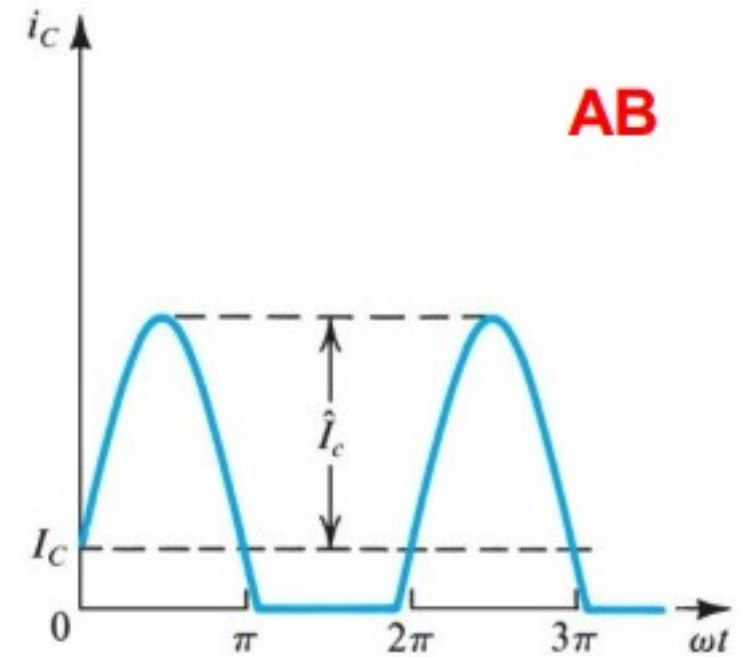
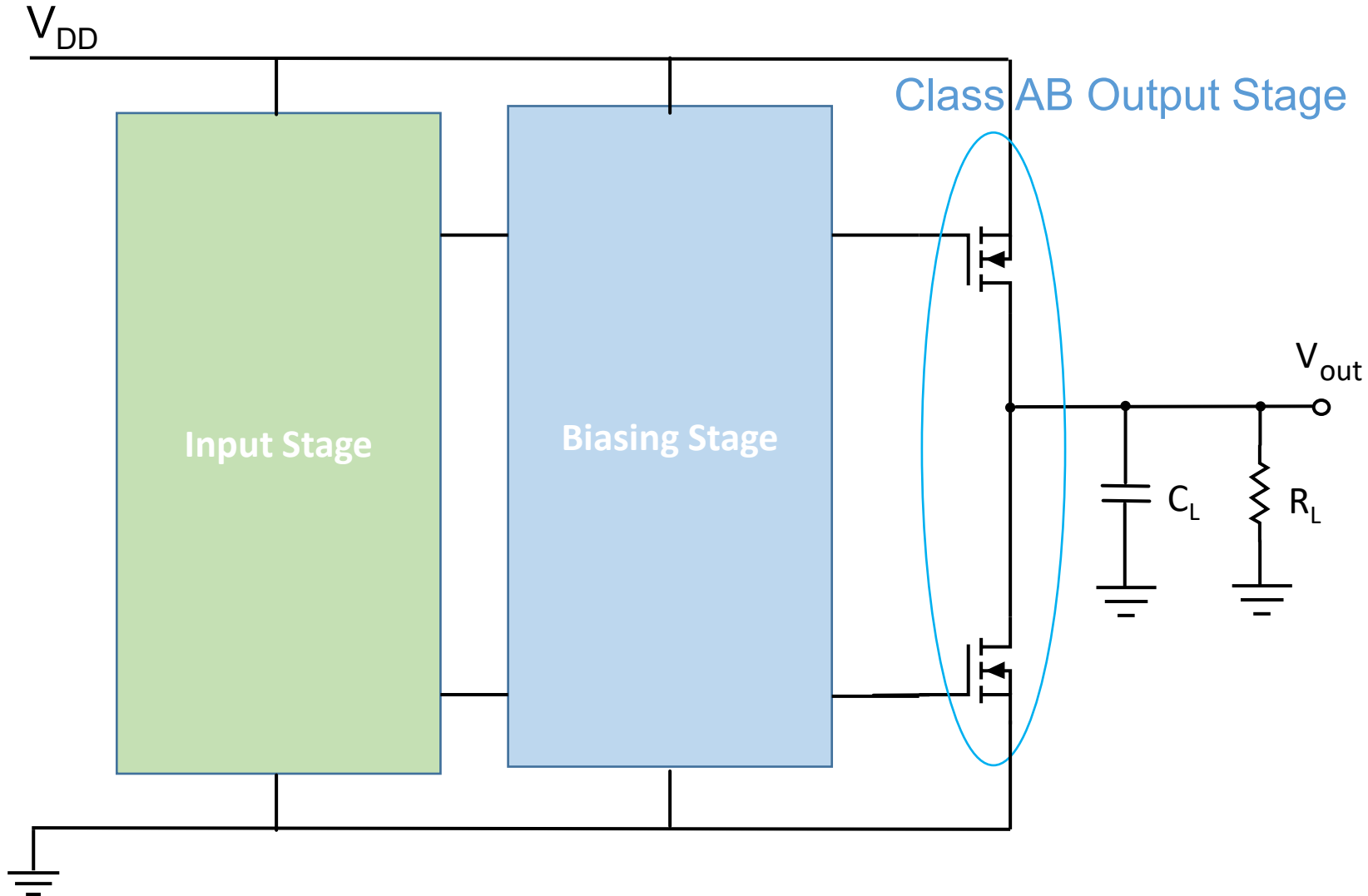


$$Efficiency = \frac{P_{RL}}{P_{VDD}} = \frac{\frac{v_{out(peak)}^2}{2R_L}}{(V_{DD})^2 \frac{1}{2} \left(\frac{2v_{out(peak)}}{\pi R_L} \right)} = \frac{\pi v_{out(peak)}}{2 V_{DD}}$$

If $V_{out} = V_{DD} / 2$, then Efficiency = 78.5%

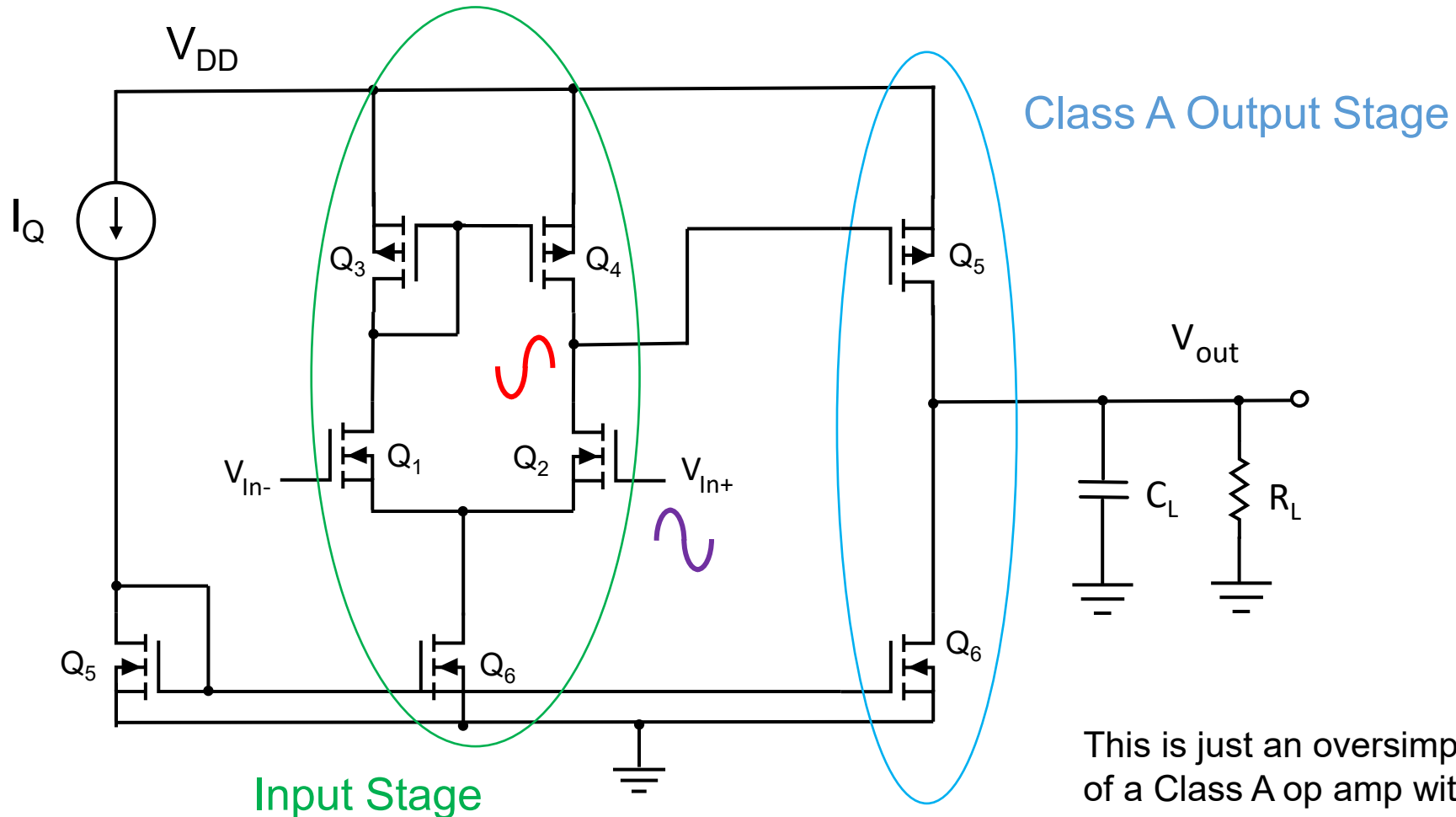
CLASS AB OUTPUT STAGE

Power efficiency of the Class-B output stage is good, but the crossover distortion causes non-linearity.



CLASS A OUTPUT STAGE IN AN OP AMP

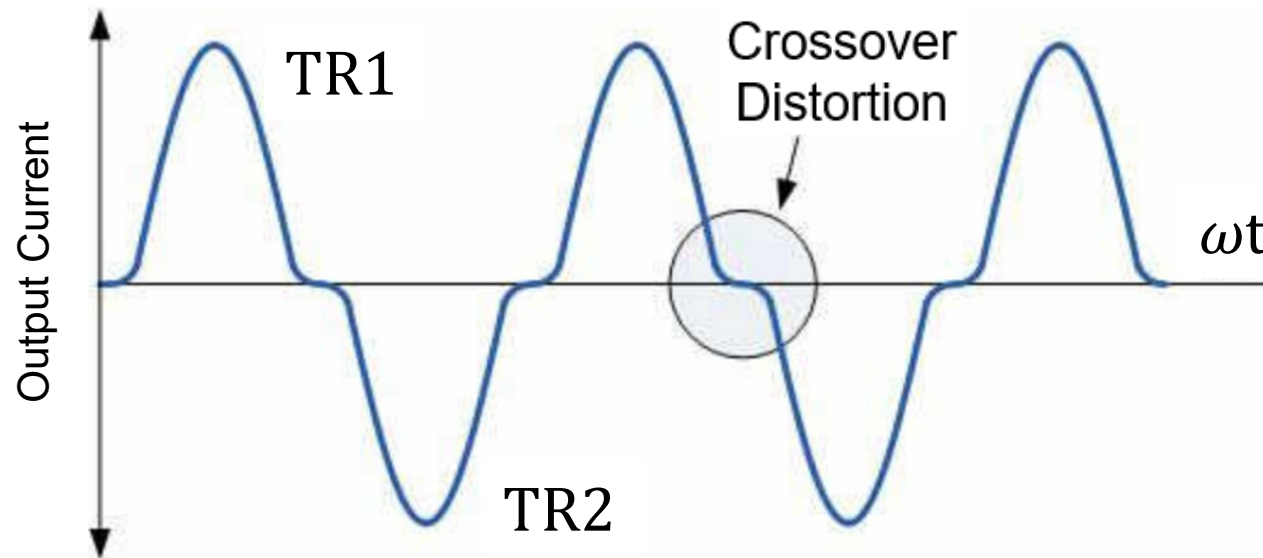
Power efficiency of the Class-B output stage is good, but the crossover distortion causes non-linearity.



This is just an oversimplified schematic of a Class A op amp without any common-mode feedback etc. Op Amps will be discussed in the next chapter...

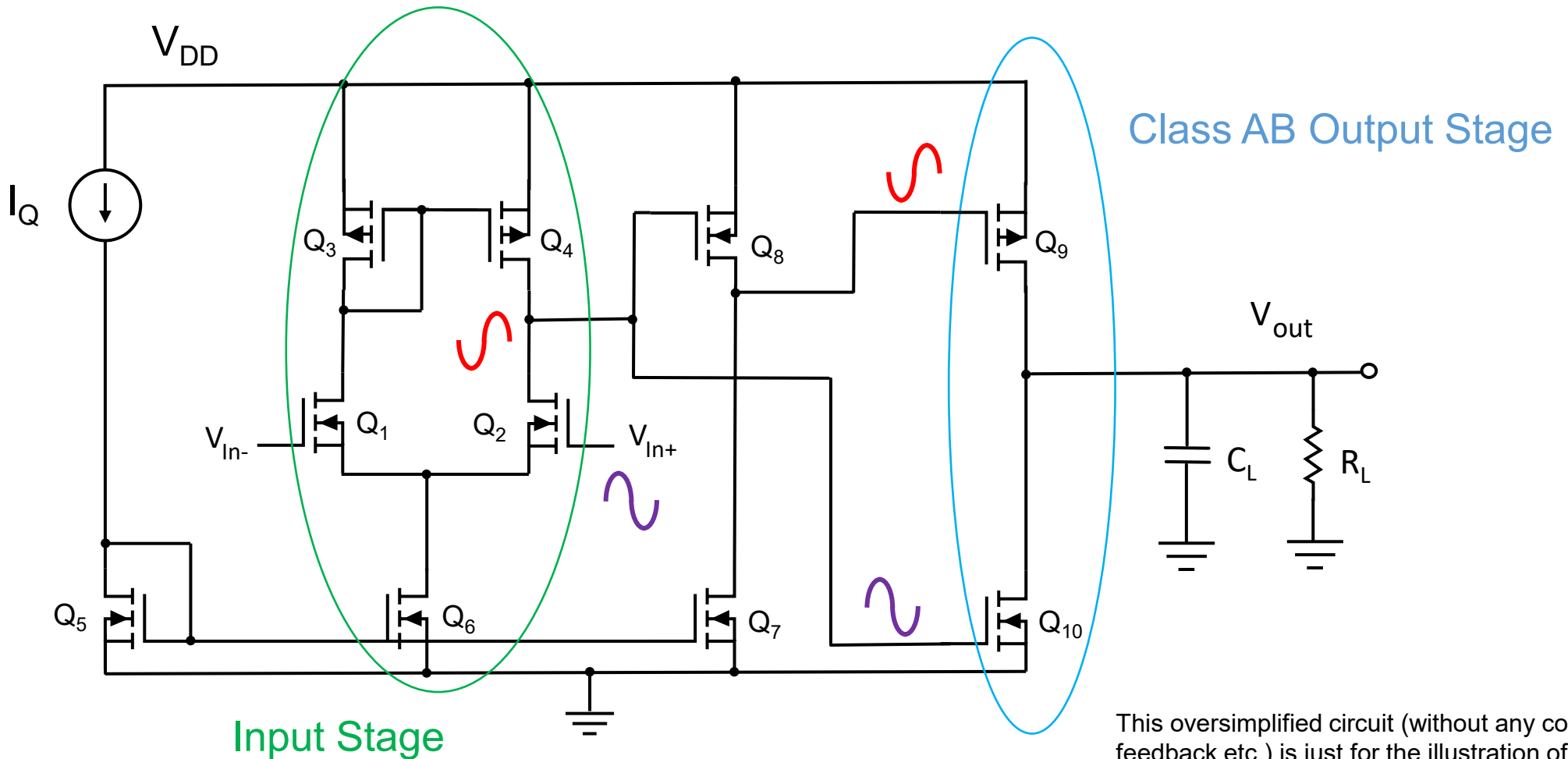
CLASS A OUTPUT STAGE IN AN OP AMP

Crossover distortion is a distortion seen in push-pull amplifiers, especially Class-B stages, when the signal passes through zero. Near this point, one transistor turns off before the other turns on because a threshold voltage is required for conduction. During this brief moment, neither transistor conducts, so the output does not smoothly follow the input, creating a small notch in the waveform. This distortion is usually reduced by using Class-AB biasing, where both transistors conduct slightly around the zero crossing.



CLASS AB OUTPUT STAGE IN AN OP AMP

Power efficiency of the Class-B output stage is good, but the crossover distortion causes non-linearity.



This oversimplified circuit (without any common-mode feedback etc.) is just for the illustration of an application of Class AB stages in op amps. Op Amps will be discussed in the next chapter...

References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.